DISPLAY REFERENCE

PowerGlove Vision Matrix Display Guide

Recognize animations, mode letters, pairing, and startup feedback.

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The blue LED matrix on the UNO Q is PowerGlove Vision's status display. It tells you which mode is active and helps distinguish startup, practice, tracking, and pairing. It does not show a game's score or confirm that a game accepted a button.

This guide describes the current Arduino sketch. Photos of the physical display will be added later; the descriptions below are usable without them.

Recognize the display

What you see	What it means	What to do
Board boot logo or system animation	The UNO Q's system software is starting, before PowerGlove Vision controls the display.	Wait for the app's hourglass or normal display.
Pulsing hourglass	The sketch has started and is waiting for the app, or the app is loading vision resources.	Allow startup to finish. If it persists, check Dashboard.
Glove animation with moving cuff, curling fingers, and a spark	Gestures are off; the app is in its idle mode.	Open Glove Academy to practice, or choose a game profile on Dashboard.
A large scanning **L**	Glove Academy lessons are active. L stands for lessons.	Practice the moves shown in your browser; controller output is paused.
A large scanning **T**	Gesture tuning is active, including hand setup.	Follow the recording, preview, and save instructions in Glove Academy. Controller output is paused.
A steady **A-I**, **BS**, or **GB**	A game profile is selected, but a calibrated hand is not currently being reported as tracked.	Show your hand and check tracking/calibration on Dashboard.
A-I, **BS**, or **GB** gently changing brightness	The app reports a detected, calibrated hand for that profile.	Check Dashboard's controller status before playing. This pulse alone does not mean controls are enabled.
ID, characters, **PN**, and digits repeating	Secure pairing is showing the device identity and temporary PIN.	Follow the Connection page; read each group in order.
A flashing **X**	The app has requested an error display.	Read Dashboard's error message before deciding whether to reconnect the camera or restart.
A blank matrix	The display has been turned off, the app is stopping, or the board has not yet reached the sketch's display stage. It may also have lost power.	Use the browser and board power indicators to distinguish these cases. Blank does not prove shutdown is complete.

Startup: logo, hourglass, then your mode

A typical startup with **Gestures off** selected is:

1. The board shows its system boot display.
2. When the Arduino sketch starts, it draws an hourglass immediately.
3. The hourglass keeps moving while the sketch connects to Linux and waits for app status.
4. Once the app reports idle mode, the glove animation begins.

With an active startup profile, the later display can instead be its profile letters. Opening Glove Academy selects **L**; enabling tuning selects **T**. These mode displays can appear while their camera resources are still starting. Use the browser's startup message for that detail.

The hourglass has five frames, changing about every 220 milliseconds. It is a repeating activity indicator, not a percentage, countdown, or promise that startup will complete. Background library preloading can continue after the app becomes idle and shows the glove. An hourglass is not required to stay visible throughout that background work.

The system controls the interval before the sketch starts. PowerGlove Vision cannot show its hourglass during that earlier interval. Cold-boot transitions still need confirmation on the physical board. If the hourglass never gives way to a mode display, open Dashboard and check whether the app is reachable and what startup or error message it reports. See startup diagnostics.

The idle glove show

The idle animation is a small repeating attraction sequence, not a reading of your actual hand. It contains these beats:

1. An energy streak crosses the matrix.
2. A cuff slides into position.
3. The glove rises into view.
4. Its fingers curl, clench, and reopen.
5. A spark travels along the glove with a short trailing glow.
6. The outline brightens, settles, and pauses before the sequence repeats.

A flashing spark here does **not** mean you performed Glove Zap. This animation means gestures are off. It is different from selecting a game profile and merely pressing **Stop controller**: that keeps the camera/profile active and can leave profile letters visible.

Glove Academy: L and T

Both letters use the same effect: a dim letter, a brighter row scanning through it, and a short glow behind that row. The scan advances about every 160 milliseconds. It is an activity animation, not a lesson-completion meter or a recording timer.

L identifies ordinary lessons, even though the navigation section is now called **Glove Academy**. **T** identifies tuning, including **Set up my hand**, individual adjustments, and previewing thresholds. The browser shows which of the three recordings you are on and whether a suggestion is ready to save.

Both modes pause controller output. After practice, use Dashboard to check the selected profile and controller state; after tuning, explicitly start controller delivery when ready to play. A pairing display can temporarily cover either letter. Tracking details and pose failures remain in the browser rather than being spelled out on the matrix.

Game profile letters

Letters	Selected profile
A through **I**	The corresponding reusable Program A-I profile
BS	Bad Street Brawler
GB	Super Glove Ball

For example, **GB** is what you should expect with Super Glove Ball selected. It stays steady when the app is not reporting both hand detection and completed calibration. It alternates between bright and dim about every 360 milliseconds when both are reported. The characters remain the same: the animation does not identify individual finger curls, movements, or button presses.

The important distinction is **tracking versus delivery**. A pulsing **GB** can appear while controller output is stopped. Confirm **Start controller** has been used and Dashboard shows output enabled, then confirm the action in the game. The matrix does not acknowledge RetroPie receipt or the game's response.

A generic **PG** ready symbol or a small pulsing tracking symbol can appear when no recognized profile identifier is available. These are fallback displays; normal named game profiles use their letters. They do not represent extra games or new gesture commands. If you expected **GB** or **BS**, check the selected profile on Dashboard.

Pairing: ID and PN

During secure pairing, the small display presents the information in pieces:

1. **ID** announces the device certificate identity.
2. Seven hexadecimal characters follow in groups of two, with the last character shown alone.
3. **PN** announces the temporary six-digit PIN.
4. Three pairs of digits follow, preserving leading zeroes.
5. The sequence repeats so you can read it again.

Each step lasts about 650 milliseconds; the nine-step loop takes roughly six seconds. Read left to right across each pair, then concatenate the groups. Do not mistake **PN** for a game program or use the certificate identity as the PIN.

Use the Connection page's certificate-comparison and PIN instructions. The pairing sequence temporarily takes priority over ordinary mode updates for the pairing display window (normally two minutes), then normal status can resume. A shutdown/off request can clear it sooner. Readability of this sequence does not prove pairing succeeded; check the Connection page's result.

For public photographs, use a clearly marked demonstration sequence or obscure live pairing digits. Never publish an active PIN or connection credentials. See [secure pairing] for the full procedure.

Errors, pauses, and a blank display

The error **X** alternates with a blank frame about every 420 milliseconds. That intentional dark half of its blink is different from a continuously blank matrix. The X does not encode a specific fault: Dashboard provides the message. Some modes prioritize their own letter or idle animation, so the absence of an X is not proof that no camera or startup error exists.

A blank display alone does not prove the board is safe to unplug. Use the normal Shutdown procedure and its completion guidance. If the board should be running, check its power, wait for boot to finish, and try Dashboard. If the web app works but the display stays blank, record what appeared immediately before it went blank and whether restarting the app changes it.

Photo examples to add later

The guide is complete as a text reference. These are useful future examples; there are no missing-image links to fill before using it.

Example	What to capture	Suggested caption
System boot	The board's logo before the app display	System startup, before PowerGlove Vision takes over
Startup hourglass	A short clip, or two successive frames	PowerGlove Vision is starting
Idle glove	Open glove, clenched glove, and spark frames	Gestures off: the idle glove animation
Academy lessons	L with its brighter scanning row	Glove Academy lessons; controller output paused
Tuning	T with its brighter scanning row	Personal tuning; follow the browser's recording steps
Super Glove Ball	Steady GB and a clip of pulsing GB	GB identifies the profile; the pulse indicates tracking
Programs and Brawler	One program letter and BS	Selected program or game profile
Pairing	ID and PN labels with synthetic or obscured values	Pairing information arrives in successive groups
Error	X plus the associated non-sensitive Dashboard message	Check Dashboard for the cause

Short clips show scanning and pulsing more clearly than a single photo. Keep the matrix upright, reduce exposure enough to separate adjacent LEDs, and include a little of the case for orientation. Record the selected mode and whether controller output was enabled. Describe photos as examples from the tested board, since camera exposure can change the apparent brightness and shape.